SECURITY ENGINEERING — HANDS-ON LAB

Automated Sign-In Risk Detection & Response

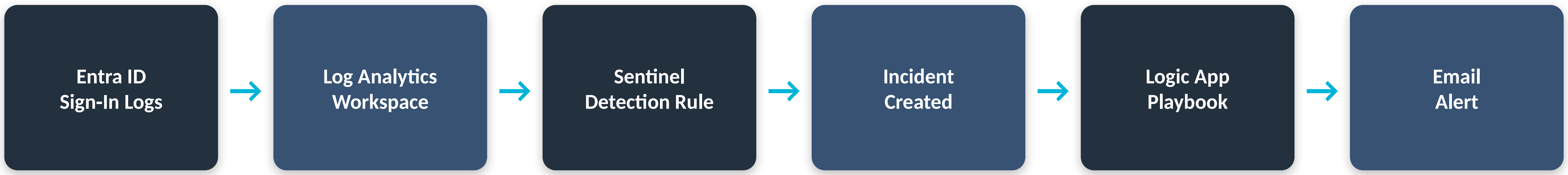
Microsoft Entra ID → Microsoft Sentinel → Logic Apps

David Rufino Botte | Senior Desktop Engineer, NZX Limited

github.com/davidrufinobotte/entra-sentinel-detection-lab

HOW IT WORKS

End-to-end pipeline: log to alert



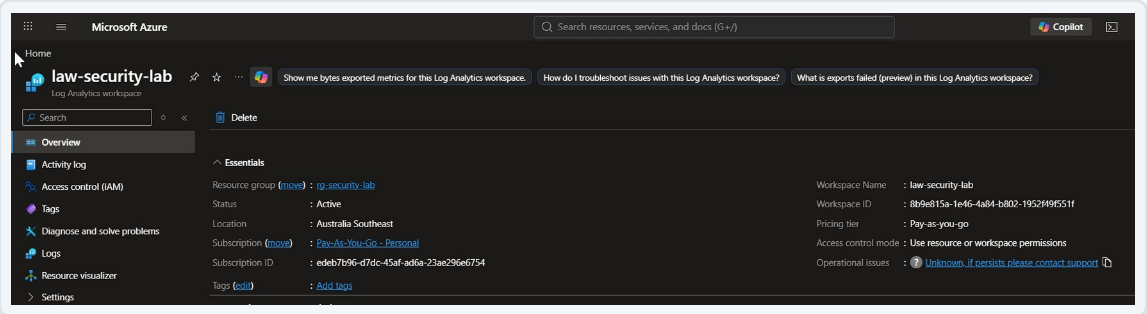
A risky sign-in is logged, ingested into Sentinel, evaluated by a scheduled detection rule, and — if it matches — automatically opens an incident and triggers an emailed alert.

01

FOUNDATION

Log Analytics Workspace

- Created law-security-lab as the central log repository for the lab
- Pay-As-You-Go subscription, Australia Southeast region
- Base layer every downstream Sentinel component reads from



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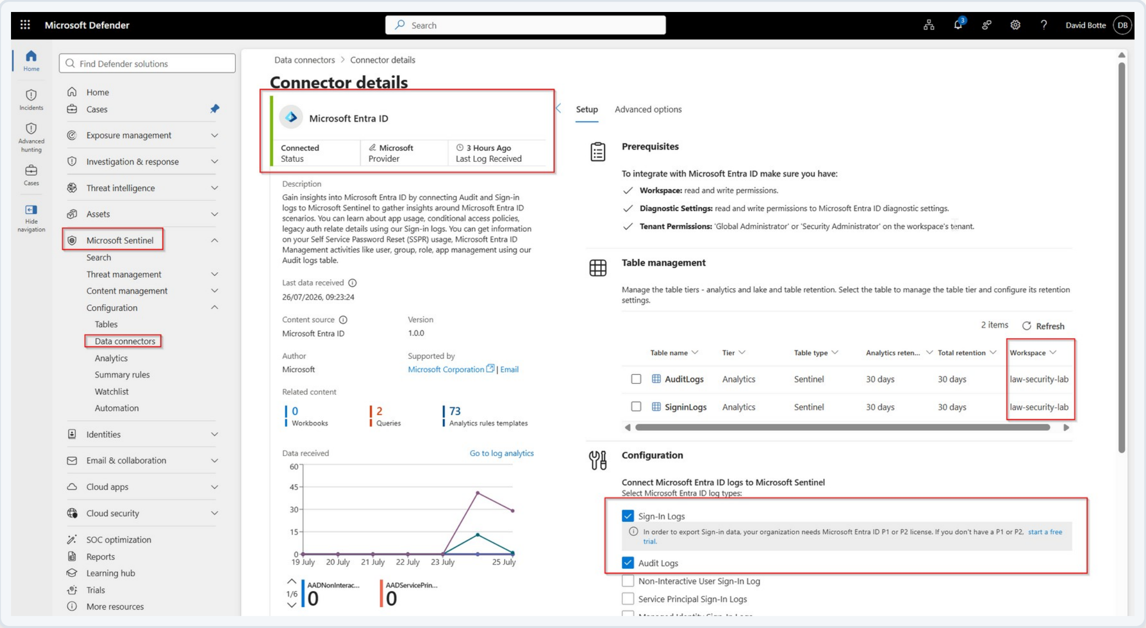
DATA SOURCE

Connecting Identity Signals



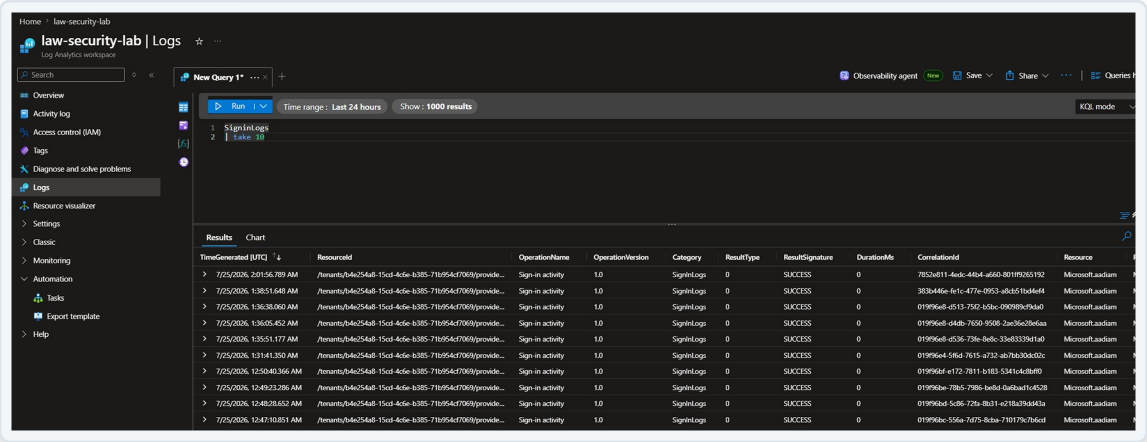
- Connected the Microsoft Entra ID data connector (Sign-In + Audit logs)
- Fixed a real issue: the connector was missing until the Entra ID solution was installed from the Sentinel Content Hub
- Confirms troubleshooting ability, not just following steps

03

VALIDATION

Confirming Data Ingestion

- Ran a KQL query (SigninLogs | take 10) before building any detection logic
- Verified logs were flowing end-to-end from Entra ID into the workspace
- Build on confirmed data, not assumptions



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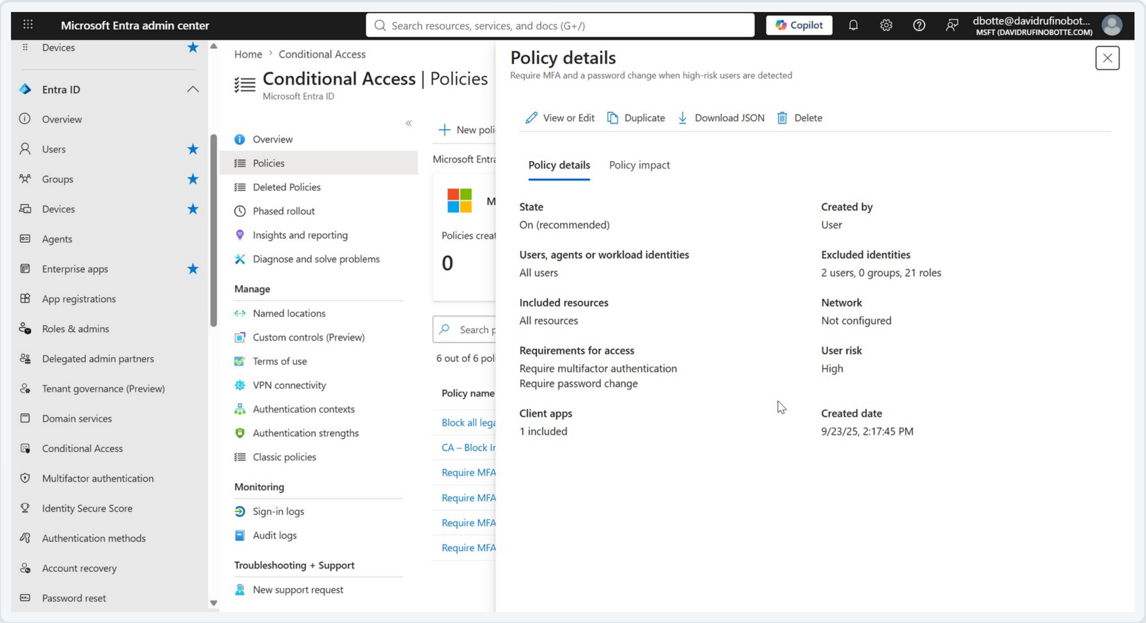
GOVERNANCE

Reusing Existing Controls



- Audited existing Conditional Access policies before creating a new one
- Found and reused a policy already enforcing MFA on risky sign-ins
- Avoids policy sprawl — a real security engineering judgment call

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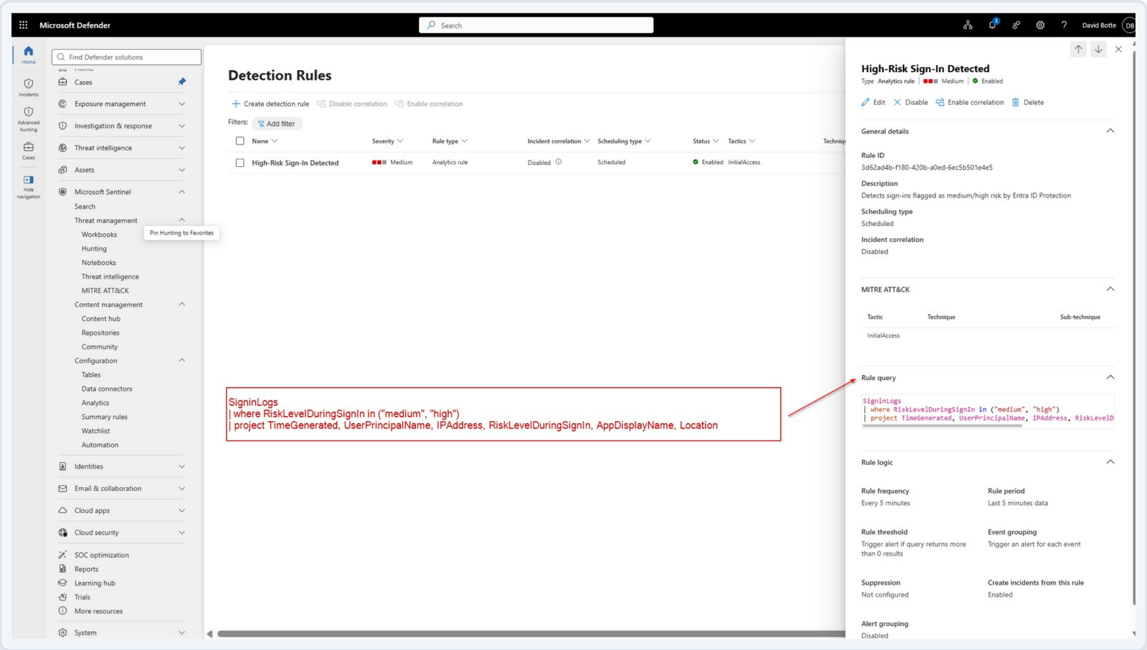
DETECTION

Building the Analytics Rule

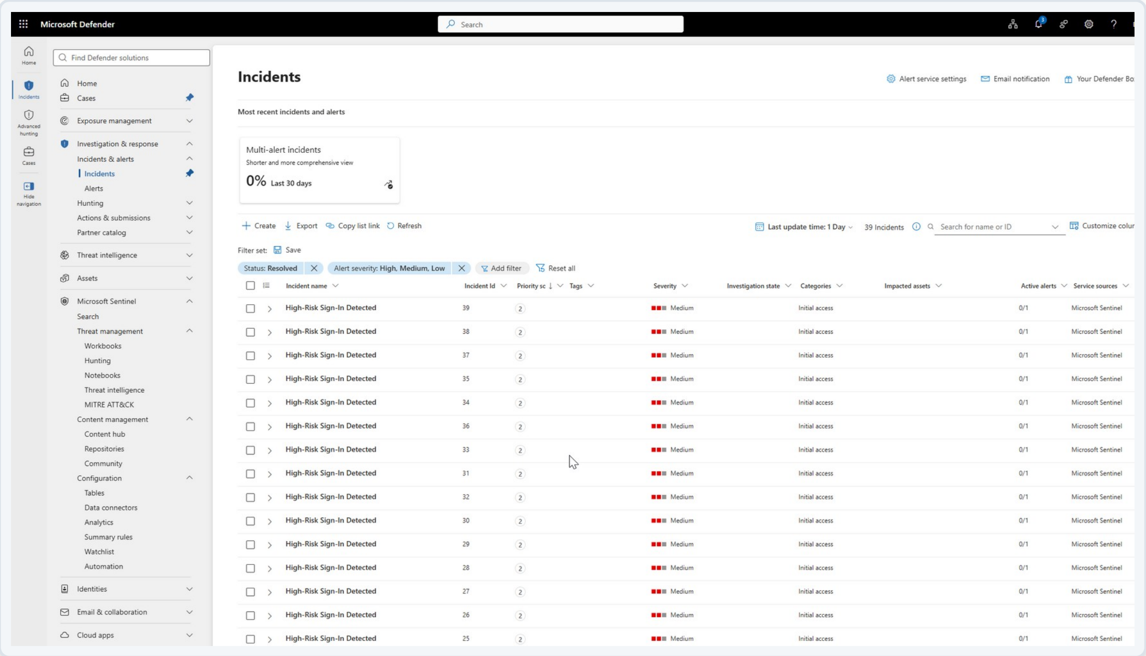
- Scheduled KQL rule 'High-Risk Sign-In Detected', running every 5 minutes
- Filters on RiskLevelDuringSignIn (medium / high) from Entra ID Protection
- Mapped to MITRE ATT&CK — Initial Access



RESULTS

06 Detections at Scale

- The rule generated 39 incidents from real sign-in activity
- Confirms the detection logic works against live data, not just theory
- Each incident automatically linked back to its source alert



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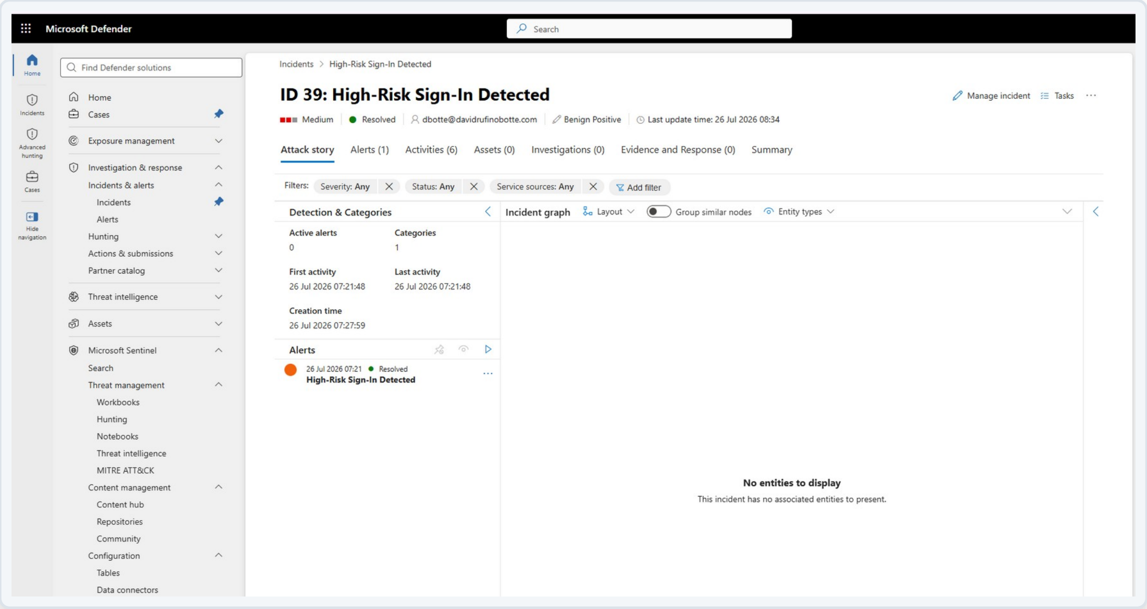
TRIAGE

Incident Review & Closure

- Reviewed and closed incidents individually with correct classification
- Classified as Benign Positive — the detection was correct; the activity was a controlled test
- Standard SOC discipline, not blanket dismissal



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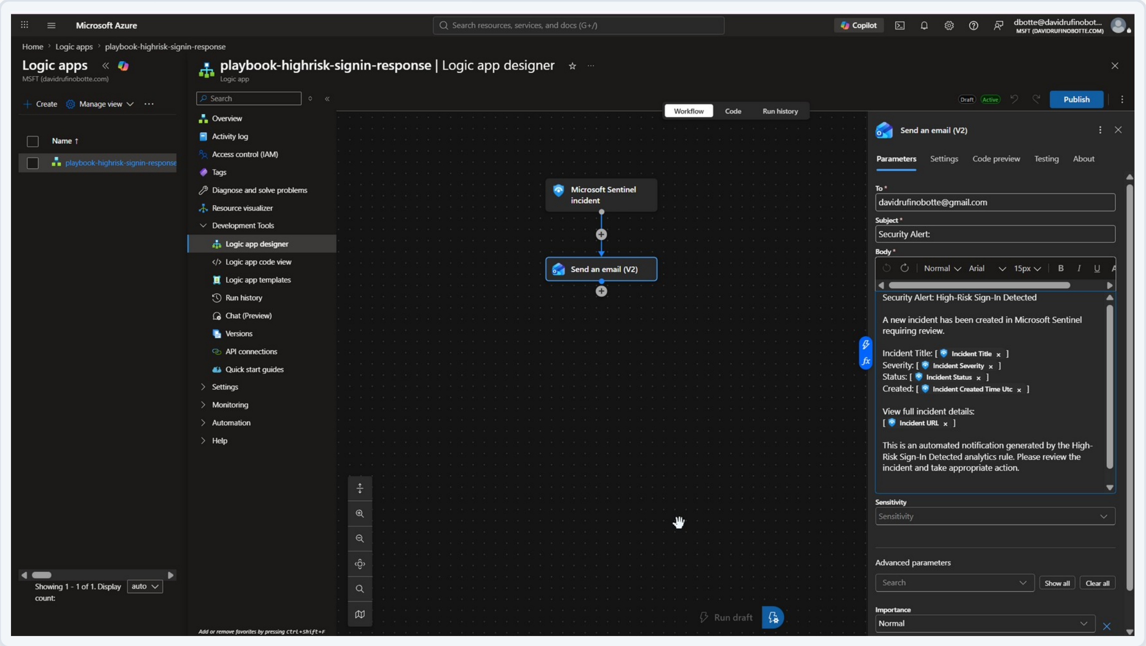
AUTOMATION

Automated Response Playbook



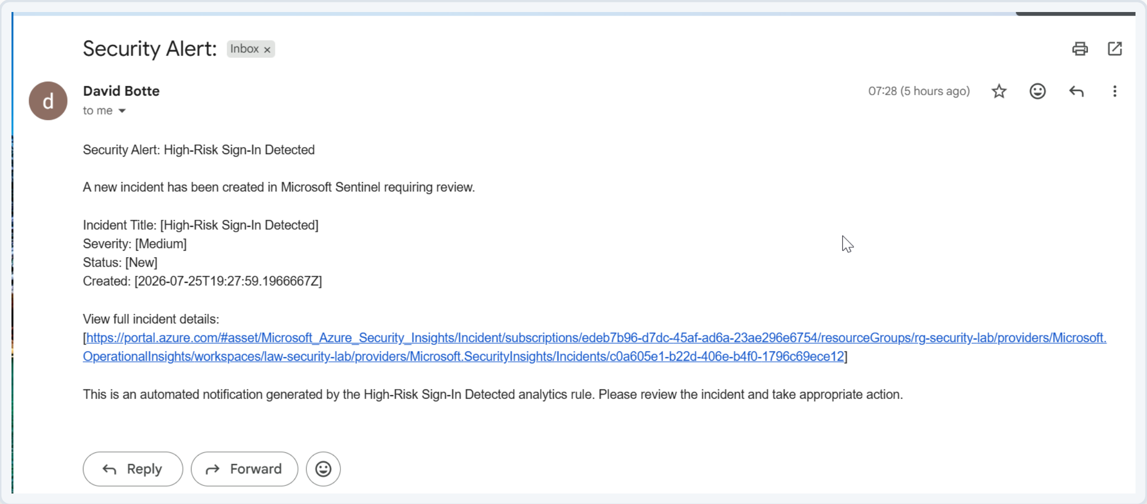
- Logic App triggered directly by 'Microsoft Sentinel incident'
- Sends a structured alert with incident title, severity, status and link
- Consumption plan used deliberately, to avoid fixed hosting cost

09

OUTCOME

Closing the Loop

- Alert delivered by email within minutes of incident creation
- Full pipeline proven: sign-in log → detection → incident → automated notification
- No manual step between detection and alert delivery



WHAT THIS DEMONSTRATES

Skills applied in this lab

- **Identity Security**

  Entra ID, Conditional Access, risk-based sign-in policy

- **Security Automation**

  Logic Apps / SOAR-style automated incident response

- **Troubleshooting**

  Diagnosed and resolved real platform issues (missing connector, permissions, cross-tenant setup)

- **SIEM Engineering**

  Microsoft Sentinel, KQL detection rules, MITRE ATT&CK mapping

- **SOC Discipline**

  Incident triage, classification, and closure — not just detection

- **Cost & Governance Awareness**

  Reused existing controls; chose consumption pricing to avoid unnecessary cost

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